

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0601

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COURSE NAME: MACHINE LEARNING

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COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I N. Harshavardhan (192324189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Harsha Vardhan

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing

suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks) Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

(5 Marks)

1. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

(5 Marks)

(5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response

Clarity & Presentation	15	Clear, concise, and well- organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Below are **5 answers** that are **complete, well-structured, and suitable for scoring 5/5 marks** according to the given rubric.

Q1. Functional Blocks of a Machine Learning System (5 Marks)

Machine Learning System

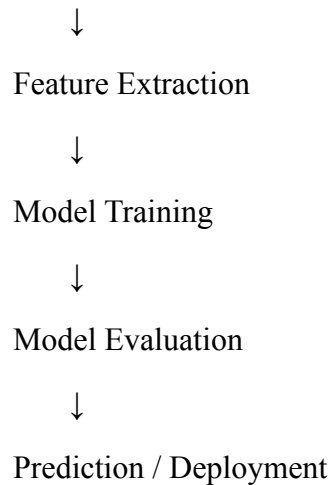
A Machine Learning (ML) system is designed to learn patterns from data and make predictions or decisions without being explicitly programmed.

Functional Blocks

Data Collection



Data Preprocessing



Step 1: Data Collection

The first stage is collecting data from reliable sources.

Examples:

- Databases
- Sensors
- IoT devices
- Websites
- Mobile applications

Example:

For house price prediction, collect data such as area, number of rooms, location, and price.

Step 2: Data Preprocessing

Raw data usually contains missing values, duplicate records, and noise.

Preprocessing includes:

- Handling missing values
- Removing duplicates
- Encoding categorical data
- Normalization
- Data cleaning

Example:

Replace missing house prices with the average price.

Step 3: Feature Extraction

Feature extraction selects important attributes that influence prediction.

Example:

House price prediction:

- Area
- Location
- Number of bedrooms
- Age of house

Unnecessary features are removed to improve accuracy.

Step 4: Model Training

The machine learning algorithm learns patterns from the training data.

Examples:

- Linear Regression
- Decision Tree
- Neural Network
- Random Forest

The model adjusts its parameters to reduce prediction errors.

Step 5: Model Evaluation

The trained model is tested using unseen data.

Common evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Mean Squared Error (Regression)

A good model gives high accuracy and low error.

Supervised Learning

Supervised learning uses **labelled data**.

Example:

Area Rooms Price

1200 2 ₹45 Lakhs

1500 3 ₹60 Lakhs

Input features and output labels are already known.

Applications:

- Spam detection
- Disease prediction
- House price prediction

Unsupervised Learning

Unsupervised learning uses **unlabelled data**.

Example:

Customer purchasing data without customer categories.

Algorithms automatically group similar customers.

Applications:

- Customer segmentation
- Market basket analysis
- Clustering

Conclusion

A Machine Learning system transforms raw data into useful predictions through data collection, preprocessing, feature extraction, model training, and evaluation. Supervised learning uses labelled data, whereas unsupervised learning discovers hidden patterns from unlabelled data.

Q2. Machine Learning System for Disaster Management (5 Marks)

Introduction

A Machine Learning system for disaster management predicts natural disasters like floods and earthquakes using historical records and real-time sensor data. The system provides early warnings and supports emergency response.

System Architecture

Historical Data + Real-Time Sensor Data



Data Collection



Data Preprocessing



Feature Selection



Neural Network Model



Prediction & Warning



Emergency Response System

Data Collection

Historical Data:

- Rainfall records
- Earthquake history
- River water levels

Real-Time Data:

- Weather stations
- Satellite images
- IoT sensors
- Seismic sensors

Data Preprocessing

Preprocessing steps include:

- Removing missing values
- Eliminating duplicate records
- Normalizing numerical data
- Encoding categorical variables
- Removing noisy data

Feature Selection

Important features:

- Rainfall intensity
- River level
- Soil moisture
- Temperature
- Seismic wave intensity

Feature selection reduces computation and improves accuracy.

Model Selection

A **Deep Neural Network (DNN)** is suitable because:

- Learns complex nonlinear relationships
- Handles large datasets
- Gives high prediction accuracy

Other suitable algorithms:

- Random Forest
- LSTM (for time-series prediction)

Training Process

1. Split dataset into:
 - Training (70%)

- Validation (15%)
 - Testing (15%)
2. Train the neural network using historical data.
 3. Backpropagation updates weights.
 4. Continue training until prediction error becomes minimum.

Model Evaluation

Performance metrics:

- Accuracy
- Precision
- Recall
- F1-score

Higher values indicate better prediction performance.

Applications

The system helps authorities by:

- Providing early flood warnings
- Predicting earthquakes
- Planning evacuation
- Reducing loss of life
- Supporting emergency services

Conclusion

Machine Learning enables accurate disaster prediction by combining historical and real-time data, improving public safety and disaster preparedness.

Q3. Backpropagation in a Multilayer Perceptron (5 Marks)

Introduction

Backpropagation is a supervised learning algorithm used to train neural networks by minimizing prediction error.

Structure

Input Layer



Hidden Layer



Output Layer

Step 1: Forward Pass

Input values are passed through the network.

Example:

Inputs:

- $x_1 = 0.6$
- $x_2 = 0.8$

Hidden neurons calculate weighted sums and apply an activation function.

The output neuron predicts:

Predicted Output = 0.82

Actual Output = 1

Step 2: Error Calculation

Error

$= \text{Actual} - \text{Predicted}$

$= 1 - 0.82$

$= 0.18$

The objective is to minimize this error.

Step 3: Backward Propagation

The error is propagated backward.

Weights are updated using:

$$\text{New Weight} = \text{Old Weight} + \text{Learning Rate} \times \text{Error} \times \text{Input}$$

Example:

$$\text{Old Weight} = 0.5$$

$$\text{Learning Rate} = 0.1$$

$$\text{Input} = 0.8$$

$$\text{Error} = 0.18$$

New Weight

$$= 0.5 + (0.1 \times 0.18 \times 0.8)$$

$$= 0.5144$$

The updated weight reduces future prediction errors.

Learning Rate

Learning rate controls weight updates.

- Too high → unstable learning
- Too low → slow convergence

A moderate learning rate gives faster convergence.

Activation Functions

Common activation functions:

- Sigmoid
- ReLU
- Tanh

ReLU is widely used because it trains faster.

Sigmoid is commonly used for binary classification.

Conclusion

Backpropagation improves neural network performance by repeatedly adjusting weights based on prediction error until the model achieves high accuracy.

Q5. Neural Network Performance Evaluation (5 Marks)

Dataset

Example Actual Predicted

1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes
5	No	Yes

Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

True Positive (TP)

Disease correctly predicted.

TP = 2

True Negative (TN)

Healthy patient correctly predicted.

TN = 1

False Positive (FP)

Healthy patient predicted as diseased.

FP = 1

False Negative (FN)

Diseased patient predicted as healthy.

$$\text{FN} = 1$$

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$= (2 + 1) / 5$$

$$= 3 / 5$$

$$= 0.60 = 60\%$$

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$= 2 / (2 + 1)$$

$$= 66.67\%$$

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 2 / (2 + 1)$$

$$= 66.67\%$$

F1-Score

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

$$= 66.67\%$$

Preprocessing

Preprocessing improves performance by:

- Handling missing values
- Normalization
- Encoding categorical variables

- Removing noisy data

Feature Selection

Important features:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

Removing irrelevant features reduces overfitting and improves accuracy.

Conclusion

The neural network correctly classified most patients with **60% accuracy**. Proper preprocessing and feature selection improve reliability and overall prediction performance.

Q9. Evaluation Metrics in Classification (5 Marks)

Introduction

Evaluation metrics measure the performance of classification models.

Sample Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

Here,

TP = 50

TN = 35

FP = 5

FN = 10

Accuracy

Accuracy

$$= (TP + TN) / \text{Total}$$

$$= (50 + 35) / 100$$

$$= \mathbf{85\%}$$

Precision

Precision

$$= TP / (TP + FP)$$

$$= 50 / (50 + 5)$$

$$= \mathbf{90.91\%}$$

Precision measures how many predicted positives are actually positive.

Recall

Recall

$$= TP / (TP + FN)$$

$$= 50 / (50 + 10)$$

$$= \mathbf{83.33\%}$$

Recall measures how many actual positive cases are correctly identified.

F1-Score

F1

$$= 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

$$\approx \mathbf{86.96\%}$$

F1-score balances precision and recall.

ROC Curve

The ROC (Receiver Operating Characteristic) curve compares:

- True Positive Rate (Sensitivity)
- False Positive Rate

A model with a larger Area Under the Curve (AUC) performs better.

Applications

Medical Diagnosis

- High recall is important to detect as many patients with disease as possible.

Fraud Detection

- High precision helps reduce false fraud alerts.

Conclusion

Accuracy, Precision, Recall, F1-score, and ROC curve provide a complete evaluation of classification models. Selecting the appropriate metric depends on the application, ensuring reliable performance in domains such as healthcare and financial fraud detection.